MIL-E-1/1632A
30 January 1975
SUPERSEDING
MIL-E-1/1632(NAVY)
29 June 1967

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6BZ6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, semi-remote cutoff
Outline --- 5-2 (EIA)
Base --- E7-1
Envelope --- T-5-1/2
Cathode --- Coated unipotential

Base connections:

Pin No.	1	2	3	4	5	6	7
Element	g1	k	h	h	a	g2	g3, int sd

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rg1	Pp	Pg2
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	W	W
Maximum:	6.9	330	0	165	---	200	1.0	2.3	0.55
Minimum:	5.7	---	---	---	---	Note 1	Note 2	---	---
TEST CONDITIONS:	6.3	125	-1.0	125	0	0	---	---	---

GENERAL:

Qualification - Required
Reliable tube

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS						UNIT
						MIN	LAL	BOGEY	UAL	MAX	ALD	
	Quality conformance inspection, part 1											
1266	Total grid current	Ec1 = -2 Vdc (see note 3)	0.65	II	Ic1	0	---	---	---	-1.5	---	µAdc
1306	Transconductance (1)		---	---	Sm	---	7,700	9,200	10,700	---	---	µmhos
1306	Transconductance (1)		0.65	II	Sm	7,000	---	---	---	12,500	---	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---	---	---	---	---
	Quality conformance inspection, part 2											
1336	Heater-cathode leakage		4.0	II	Ihk	---	---	---	---	10	---	µAdc
1301	Heater current		4.0	Code G	If	275	286	---	314	325	---	mA
1256	Electrode current (anode)		4.0	Code G	Ib	10.0	13.0	17.0	21.0	25.0	---	mAdc
1256	Electrode current (screen grid)		4.0	Code G	Ic2	3.0	4.0	5.0	6.0	7.0	---	mAdc
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	---	---	---	18	---	%
1306	Transconductance (3)	Ec1 = -4.5 Vdc	4.0	Code G	Sm	450	---	900	---	1,700	---	µmhos
1306	Transconductance (4)	Ec1 = -19 Vdc	4.0	Code G	Sm	20	---	75	---	150	---	µmhos
1306	Transconductance (5)	Ec1 = -8 Vdc	4.0	Code G	Sm	250	---	---	---	650	---	µmhos
1211	Insulation of electrodes		4.0	Code G	---	---	---	---	---	---	---	---
1246	Audio frequency noise	Esig = 700 mVac; Ebb = 250 Vdc; Ecc2 = 250 Vdc; Rg1 = 0.1 Meg; Rg2 = 1.0 Meg; Rp = 0.2 Meg; Rk = 4,000 ohms; Ck = 25 µF (see note 4)	2.5	I	---	---	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	No external shield	4.0	Code G	Cgp	---	---	---	---	0.025	---	pF
					Cin	5.8	---	---	---	7.6	---	pF
					Cout	2.0	---	---	---	2.5	---	pF
					Chk	3.0	---	---	---	4.6	---	pF
(1)1031	Low-frequency vibration	Ec1 = -2 Vdc; Rp = 2,000 ohms	2.5	Code G	Ep	---	---	---	---	250	---	mVac
1041	Shock	300 G	---	---	---	---	---	---	---	---	---	---
1031	Vibration fatigue		6.5	Note 5	---	---	---	---	---	---	---	---
---	Post-shock and vibration fatigue test end points:											
1031	Low-frequency vibration		---	---	Ep	---	---	---	---	400	---	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	---	---	---	20	---	µAdc
1306	Change in transconductance (1) of individual tubes		---	---	ΔSm t	---	---	---	---	20	---	%

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS						UNIT
						MIN	LAL	BOGEY	UAL	MAX	ALD	
	Quality conformance inspection, part 2 -Continued											
2126	Glass strain		2.5	I	---	---	---	---	---	---	---	---
1121	Base strain	Note 6	---	---	---	---	---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---	---	---	---	---
	Quality conformance inspection, part 3											
1501	Intermittent life	Group A; Eb = 200 Vdc; Ec = 0; Ec2 = 150 Vdc; Rk = 180 ohms; Rgl = 1.0 Meg; Ehk = +150 Vdc with 117 Vac superimposed	---	---	---	---	---	---	---	---	---	---
---	Intermittent life-test end points: (500 hours)											
1266	Total grid current		---	---	Icl	0	---	---	---	-3.0	---	µAdc
1306	Transconductance (1)		---	---	Sm	6,000	---	---	---	---	---	µmhos

NOTES:

1. Maximum rating for positive dc heater cathode voltage shall be 100 Vdc.
2. For fixed bias operation, the maximum rating for Rgl shall be 0.25 megohm.
3. (A) This test to be performed at the conclusion of the holding period.
4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.
6. Not required for qualification.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2894)

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